Supplementary Material for "OmniST: Topology-Invariant Forecasting for Dynamic Sensor Networks With Guaranteed Coverage at Unseen Sensors"

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This document collects results that support the main paper but did not fit its page budget. Every number here comes from the same committed result files as the main text and reproduces from the released scripts at https://github.com/Praveg432/OmniST.

S1. Month-by-Month Accuracy Against Published E-STGCN Numbers

Table S1 lists 30-day $PM_{2.5}$ MAE for each month of 2023 on the Delhi CPCB data, next to the values published for E-STGCN [1] on the same split. The OmniST values come from a single representative run, kept for month-level readability; the seed-hardened figure in the main text is the 5-seed MAE of 28.2 ± 0.9 μg/m^3. Because the E-STGCN numbers are published values under a protocol we did not run, the controlled reimplementation in the main paper is the comparison we rely on.

TABLE S1
30-Day $PM_{2.5}$ MAE (μg/m^3) on Delhi CPCB Data (2023), Single Run

Month	OmniST	E-STGCN*	Note
Jan	57.3	56.1	Comparable
Mar	24.2	25.8	Comparable
Apr	22.0	25.4	OmniST better
May	30.9	29.0	Comparable
Jun	18.1	22.0	OmniST better
Jul	11.3	15.7	OmniST better
Aug	11.2	11.9	Comparable
Sep	15.2	15.1	Comparable
Oct	30.4	36.4	OmniST better
Nov	69.8	77.1	OmniST better
Dec	50.9	116.8†	See note
Avg	31.0	39.2	

*From E-STGCN Table 2 [1]; direct comparison requires caution.
†Exceeds naive baselines; a methodological difference is likely.

In the moderate-pollution months (April to September) the two methods are within 3 μg/m^3 of each other. In December the published E-STGCN MAE (116.8) exceeds even a naive "predict the recent average" baseline (48.9 μg/m^3). We cannot tell whether that reflects a methodological difference or a real failure mode without running the authors' code, so we make no claim of winter superiority.

S2. Ablation of the Spatial Component

Table S2 ablates the spatial attention and the neighbourhood size on coverage and inductive MAE at unseen nodes. Removing the attention entirely (coordinate MLP plus temporal backbone only) or changing k moves coverage by at most about 0.01 on every domain. The coverage guarantee is therefore earned by the coordinate-based inductive construction, which lets each node be scored from its own coordinates and local context, rather than by the attention layer or a particular k. The attention's role is to make inductive prediction possible and to sharpen intervals slightly.

S3. Calibration Diagnostics and Multiple Coverage Targets

Fig. S1 plots empirical against target coverage for $\alpha \in \{0.5, \ldots, 0.95\}$ on each dataset; every curve tracks the diagonal. A split by residual tercile shows near-complete coverage on the low and middle strata and lower coverage on the highest stratum (≈ 0.73–0.78). Because the strata are defined by residual size, the high stratum is partly hard by construction; we report it as a limit of marginal coverage rather than a separate failure.

Repeating the inductive evaluation at targets of 80, 90 and 95% (5 seeds for air quality, 3 for traffic) gives empirical coverage within about 3 points of the target in every cell: Delhi $0.824/0.915/0.958$, Beijing $0.801/0.928/0.968$, and traffic $0.773/0.911/0.963$. The loosest cell is traffic at the 80% target (0.773).

TABLE S2
ABLATION: COVERAGE AND INDUCTIVE MAE AT UNSEEN NODES (TARGET 90%)

Dataset	Metric	Full	No GAT	k=4 / k=16
Delhi AQ	coverage	0.915	0.912	0.914 / 0.913
	ind-MAE	26.7	26.4	26.2 / 26.1
Weather	coverage	0.917	0.925	0.920 / 0.922
	ind-MAE	4.99	4.80	4.92 / 4.97
Traffic	coverage	0.919	0.919	0.919 / 0.919
	ind-MAE	9.55	9.54	9.63 / 9.63

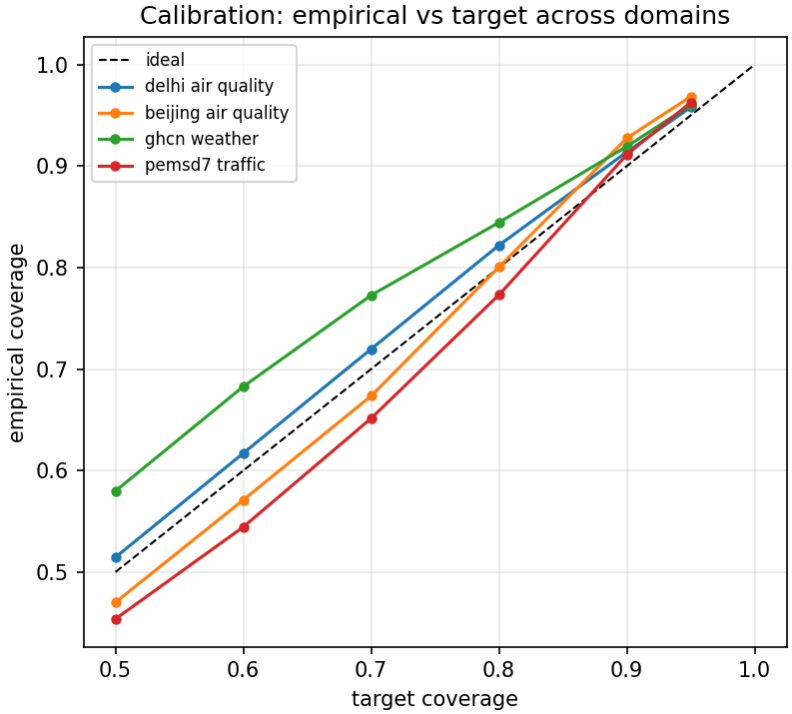


Fig. S1. Calibration curves (empirical against target coverage) across the four real datasets.

S4. CONDITIONAL COVERAGE AND MONDRIAN CONFORMAL

The guarantee is marginal. Per-unseen-node coverage at the 90% target is close to conditional on the air-quality networks (spread 0.87–0.95) but not on the irregular traffic network (spread 0.73–0.98, averaging a valid 0.92). Grouping unseen nodes by distance to the nearest training node and calibrating a separate quantile per group (Mondrian conformal) lifts the worst-group coverage on traffic from 0.80 to 0.86, giving that group a wider interval (51.5 against 42.1), and leaves the air-quality domains unchanged (Fig. S2). This narrows the gap without closing it ($0.86 < 0.90$).

S5. COMPUTATIONAL SCALING

A cost stress test that replicates real sensors with coordinate jitter shows training time growing super-linearly ($169 \to 437 \to 1106$ s for $228 \to 456 \to 912$ nodes), the signature of exact $O(N^2)$ k-NN attention. This test does not certify coverage, because duplicating nodes breaks calibration exchangeability. A scalable variant (a cKDTree neighbour index with an index-select gather) is numerically equivalent to exact attention (output cosine similarity 0.99996 at N=228) and 2.3× faster per forward pass at N=1824, with the gap widening in N.

REFERENCES

[1] M. Panja, T. Chakraborty, A. Biswas, and S. Deb, "E-stgcn: Extreme spatio-temporal graph convolutional networks for air quality forecasting," *Journal of the Royal Statistical Society Series A: Statistics in Society*, 2026.

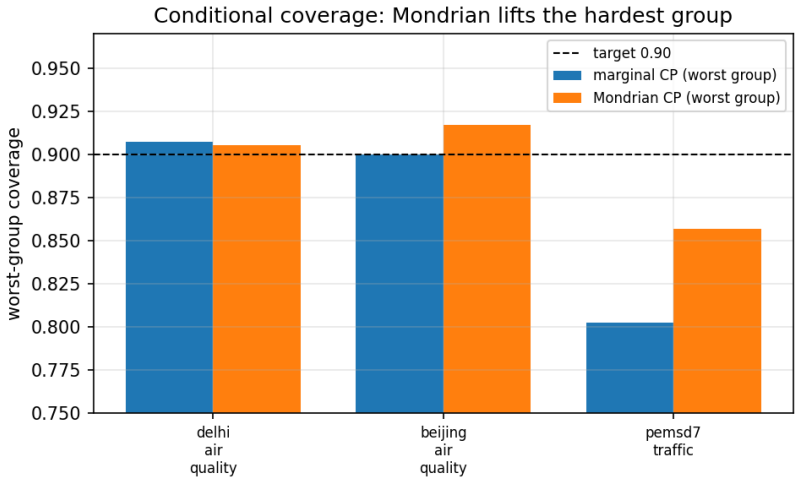


Fig. S2. Mondrian conformal lifts the worst-group coverage on the traffic network from 0.80 toward the target and leaves the air-quality domains unchanged.